



Polyaxial Stress-Dependent Tensorial Permeability Variations of a Columnar Jointed Rock Mass: Insights from 3D Distinct Element Method

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Abstract Stress-induced permeability changes in geological formations have implications for various geo-engineering applications. Fractures are the predominant fluid flow pathways within tight geological formations such as jointed basaltic rock mass. Three-dimensional permeability variations of a columnar jointed rock mass using the three-dimensional block distinct element method are investigated. First, a permeability tensor is constructed by simulating fluid flow through joint sets using cubic law and determining its principal components through eigenvector analysis. Subsequently, numerical simulations are conducted to gain insights into the impact of principal stresses on joint deformation and fluid flow. Joint deformation is controlled by the joint's mechanical properties, contact models, and stress state. Different isotropic and anisotropic stress loading scenarios on the permeability tensor evolution are investigated. The research findings indicate that isotropic stress loading led to uniform normal compression on fractures, resulting in a monotonic reduction of all permeability tensor components and increased permeability anisotropy. Anisotropic stress loading, particularly the intermediate stress, often neglected, significantly influenced the rock's overall permeability.

While the directions of the principal components remained consistent during isotropic loading, they exhibited changes in both direction and magnitude during anisotropic loading. The relative angle between the joint plane and the principal stresses was crucial in controlling joint behavior, including normal compression, shear failure, and dilation. Additionally, data analysis revealed that the exponential empirical model provided an excellent fit for permeability–stress data.

Keywords Stress-dependent permeability · Fractured rock · Distinct element method · Intermediate principal stress · Permeability tensor

Introduction

Fluid flow in rocks is pivotal to a wide array of geological, environmental, and engineering processes. Hydrogeologically, it is crucial for managing aquifers, understanding contaminant transport, and ensuring sustainable groundwater use [1]. In petroleum geology, fluid dynamics are fundamental for hydrocarbon migration and reservoir management [2]. Geotechnical applications include assessing slope stability and designing tunnels and dams [3]. Environmental considerations encompass waste disposal safety and carbon sequestration efficiency [4]. Additionally, fluid flow underpins geothermal energy extraction and certain mining techniques [5, 6].

The external tectonic stresses and pore pressure in the subsurface are dynamic properties that evolve over time. Subsurface rock properties, such as mechanical characteristics and flow behavior, are sensitive to these stress variations [7, 8]. Mechanical stresses can deform the flow pathways within the rock, thereby significantly influencing the rock's flow capacity. Pioneering studies have provided

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experimental evidence of the dependence of permeability on stress [9, 10].

Research on stress-dependent permeability can be categorized into three primary categories: laboratory studies, the development of analytical models, and numerical simulations. Initial experimental research demonstrated the phenomenon of stress-dependent permeability [9]. Subsequent investigations aimed to understand the separate influences of pore pressure and confining pressure on stress-dependent permeability, leading to the formulation of effective stress laws [10]. Further laboratory studies examined various rock types to observe how stress variations induce changes in permeability. Notable examples include studies on shales and tight rocks [11–13], crystalline and fractured rocks [14–16], and coals [17]. Field-scale studies highlighted significant reductions in initial stage hydrocarbon production rates, with reductions of up to 50% attributed to permeability decreases [18]. These studies revealed a broad range of stress-induced permeability sensitivity, with some permeabilities decreasing by as much as 98% from their original values [19]. These findings underscore the critical importance of understanding and accounting for stress-dependent permeability in subsurface rock formations to optimize hydrocarbon extraction processes.

Numerous analytical models [20] and empirical models [21, 22] have been devised to characterize the phenomenon of stress-dependent permeability. While these empirical and analytical equations align well with experimental observations, they fall short of capturing the permeability alterations brought about by shear-induced dilations and shear failures that manifest along fracture planes within complex fracture networks and heterogeneous rocks. Consequently, numerical modeling emerges as a powerful tool for comprehending the subtle fluid flow dynamics inherent to these interacting fracture networks.

Numerical simulation studies of hydromechanical coupling in fractured rocks are broadly categorized into three classes: continuum methods, discrete methods, and hybrid methods [23]. The finite element analysis method (FEM) is the most popular in continuum methods, where fractured rock is represented as a continuum with thin or zero-thickness interfaces explicitly embedded into it [24, 25]. However, continuum assumptions make it challenging to account for fracture growth, sliding block rotation, or large fracture openings, necessitating remeshing. The extended finite element methods (XFEM) improve upon traditional FEM by significantly reducing computational cost and complexity, as well as the need for remeshing [26, 27].

In discrete methods, the material is represented as an assemblage of particles or blocks, and contacts between them are assigned appropriate elastic properties and constitutive models [28]. The discrete element method has proven successful in understanding permeability evolution

under varying stresses in fractured rocks. For example, hydromechanical coupling in fractures, including normal fracture deformation, shear dilation, and shear failure, is successfully modeled using the Universal Distinct Element Code (UDEC) in two-dimensional scenarios [29, 30]. Additionally, numerical modeling on coal samples has revealed the development of anisotropic strain and permeability [31]. Several variables control the permeability evolution, including the mechanical properties of fractures such as stiffness, strength, initial and residual cohesion, and geometrical properties including fracture length, distribution, and alignment, as well as the mechanical properties of the host rock [16, 32]. These findings highlight the complex interplay between mechanical and hydraulic properties in determining the permeability of fractured rock systems.

The previous research in this field has primarily focused on two-dimensional investigations, assuming either isotropic or transversely isotropic mediums. However, there has been limited exploration of the impact of polyaxial stress on tensorial permeability variations and anisotropic deformation. The typical 2D analysis fails to reflect the full triaxial stress-dependent behavior of three-dimensional (3D) fractured rock samples because it disregards intermediate stress effects [33]. Moreover, relying on a 2D approximation of a 3D geological model may result in a substantial underestimation of overall permeability, often by an order of magnitude greater than three, and may lead to erroneous estimations of the principal permeability orientations [34]. Furthermore, it is essential to recognize that permeability is a tensorial quantity, implying that the tensor components can vary distinctly along different principal stress components.

Columnar jointed rocks are significant geological features for many practical engineering applications such as groundwater flow, hydraulic dam construction sites, and nuclear waste disposal sites. Therefore, it is essential to study the effects of stress on these geological formations [35, 36]. To date, minimal research has explored the effects of polyaxial stress on permeability within three-dimensional geological models. This paper introduces a novel investigation using the three-dimensional distinct element method to study polyaxial stress-dependent variations in the permeability tensor of columnar jointed rocks. The research methodology involves constructing and validating a three-dimensional permeability tensor against established analytical solutions. Subsequently, the one-way hydromechanical coupling problem is solved by simulating the model under various in situ stress scenarios to observe changes in the magnitude and direction of the permeability tensor components. The uniqueness of this research lies in its emphasis on polyaxial stresses, as opposed to just two-dimensional stress conditions, providing a more realistic representation of subsurface phenomena.

Methodology

Numerical simulations in this study were conducted using the three-dimensional distinct element code (3DEC), a block discrete element code [37]. The following sections detail the modeling principles and governing laws of 3DEC. To ensure the robustness of the methodology, the models were validated against existing analytical solutions and previously published data.

Distinct element method

In three-dimensional distinct element code (3DEC), a jointed rock is explicitly represented as an ensemble of distinct blocks joined together by joint planes [28, 37, 38]. The model consists of two primary components: blocks and joints, both of which are deformable based on their elastic properties (see Fig. 1). In the 3DEC, block discrete element method, the material is represented as discrete blocks, each of which can be either rigid or deformable (see Fig. 1). Discontinuities within the medium are treated as interfaces between these blocks, allowing for large displacements along the discontinuities and rotations of the blocks. Deformable blocks are further divided into a mesh of finite difference elements. Each element within this mesh responds to stress and strain according to a predefined stress–strain relationship. Additionally, the movement of discontinuities is governed by force–displacement relationships, allowing for both normal and shear displacements. Only pertinent details regarding the 3DEC are presented in the following sections. For a more comprehensive understanding of the discrete element method, readers are encouraged to refer to [39]. For specific details regarding the implementation of 3DEC, the 3DEC code manual [40] is recommended.

In the present study, joints are characterized by properties including hydraulic aperture (u_h), mechanical aperture (u_m), normal stiffness (K_n), shear stiffness (K_s), cohesion (C), friction angle (ϕ), and dilation angle (θ_d). The contact area of the asperities within the joint controls both the hydraulic and mechanical apertures [41]. For simplicity, in this study,

these two quantities, hydraulic (u_h) and mechanical apertures (u_m), were assumed to be equal. The effective stress on the joint walls, determined by the difference between the total stress and the pore pressure ($\sigma_n - P_p$), influences joint deformation. Joints undergo normal and shear deformation when subjected to normal and shear forces, respectively. Before the onset of failure, the governing equations can be expressed as follows [40]:

$$\Delta\sigma_n = K_n^{\text{current}} \Delta u_n \quad (1)$$

$$K_n^{\text{current}}(\sigma) = K_n^0 \times \sigma_n^e \quad (2)$$

similarly,

$$\Delta\sigma_s = K_s^{\text{current}} \Delta u_s \quad (3)$$

$$K_s^{\text{current}}(\sigma_s) = K_s^0 \times \sigma_s^e \quad (4)$$

Here $\Delta\sigma_n$ (Pa) represents the change in effective normal stress, $\Delta\sigma_s$ (Pa) denotes the change in effective shear stress, Δu_n (m) indicates the change in normal displacement, and Δu_s (m) is the increment in shear displacement. K_n (Pa m⁻¹) is the fracture normal stiffness, and K_s (Pa m⁻¹) is the fracture shear stiffness. K_n^0 and K_s^0 are the initial joint normal and shear stiffness, respectively. The nonlinear behavior of K_n and K_s is illustrated in Fig. 2. Normal compressive stress reduces the aperture, while normal tensile stress increases it. Similarly, joint slip occurs when the shear stress exceeds the shear strength of the joint, as governed by the Mohr–Coulomb criteria. The peak strength of the joint can be expressed as follows:

$$\tau^{\text{peak}} = C + (\sigma_n - P_p) \tan \phi \quad (5)$$

where τ^{peak} (Pa) represents the peak shear strength, C (Pa) is the cohesion, and ϕ (degrees) is the friction angle of the joints. Furthermore, once the joint starts to slip, the residual shear strength can be expressed as follows:

$$\tau^{\text{residual}} = C_{\text{res}} + (\sigma_n - P_p) \tan \phi_{\text{res}} \quad (6)$$

where C_{res} and ϕ_{res} indicate residual cohesion and residual friction angle of the joint, respectively. Dilation on fractures occurs when the rough fractures undergo shear deformation, and the dilation effect is incorporated in terms of dilation angle (θ_d), as:

$$\Delta\sigma_n = K_n^{\text{current}} \Delta u_s \tan \theta_d; \quad u_s < u_{cs} \quad (7)$$

$$\Delta\sigma_n = 0; \quad u_s > u_{cs} \quad (8)$$

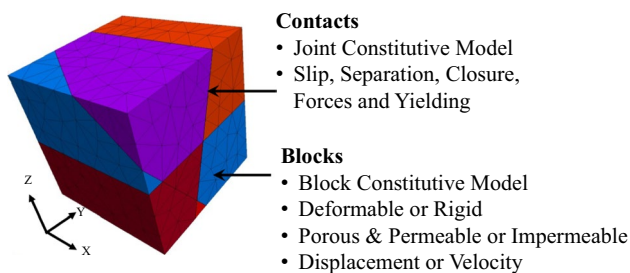
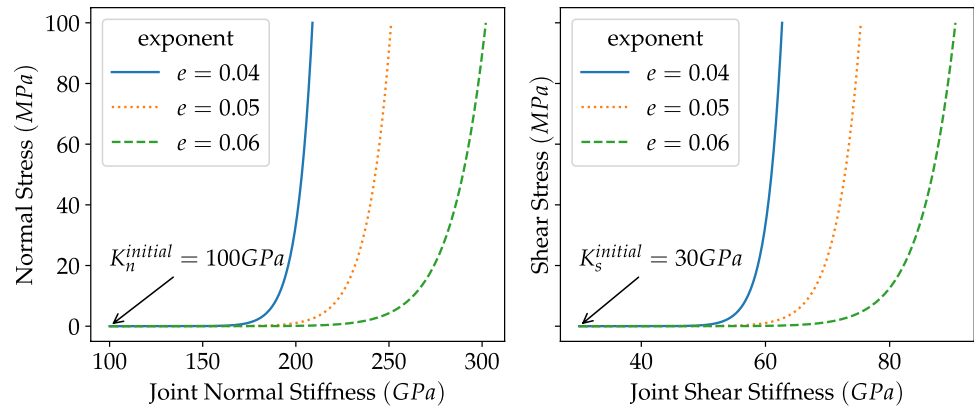


Fig. 1 A schematic of the 3DEC model showcasing its essential components: blocks and contacts (joints)

Fig. 2 The figure illustrates the variation of normal stress with joint normal stiffness (left) and the variation of shear stress with joint shear stiffness (right) at three exponent values, displaying a nonlinear increasing trend (refer to Eqs. 2 and 4)



where u_{cs} (m) is the critical non-elastic shear displacement beyond which there will be no dilation (see Fig. 5).

Fluid Flow Model

Fluid flow within rocks can occur both inside the joints and within the blocks. However, this study assumes that flow occurs only within the fractures. This assumption is consistent with certain real-world geomechanical phenomena. For example, in basaltic columnar joints and ultra-tight reservoirs, fluid predominantly flows through the fractures, while the matrix remains impermeable [42]. To simplify the analysis, we assume that the fractures are smooth-walled and parallel. Additionally, the fluid is incompressible and flows in a laminar pattern. The fluid movement within these fractures adheres to the cubic law, a simplified form of the Navier–Stokes equation, [40, 43], which can be expressed as follows:

$$q = -\frac{u_h^3}{12\mu} \frac{dP}{dx} \quad (9)$$

Here q ($\text{m}^2\text{sec}^{-1}$) represents the flow rate per unit width of the fracture, $\frac{dP}{dx}$ (Pa m^{-1}) denotes the fluid pressure gradient, and μ (Pa s) is the liquid viscosity. To meet the requirements of the cubic law equation (Eq. 9), maintaining a laminar flow of the fluid is necessary. It is important to recognize that using Eq. (9) is appropriate when the flow primarily takes place in fractures or very thin planar flow channels. However, this method becomes unsuitable if the flow occurs through large, irregularly shaped three-dimensional voids between solid blocks. Additionally, the applicability of the cubic law is determined by the Reynolds number, which, in turn, is affected by hydraulic aperture u_h . Furthermore, using Eq. (9), the absolute permeability of a single fracture can then be expressed as follows [43]:

$$k = \frac{u_h^2}{12} \quad (10)$$

where k (m^2) is the absolute permeability of a single joint, and u_h (m) is the joint's hydraulic aperture. The flow equation indicates that the joint's hydraulic aperture predominantly governs the permeability of the joints. Consequently, variations in the permeability of a jointed rock are essentially changes in the apertures of the joints. These aperture changes can result from normal displacement, shear displacement, or dilation of rough fractures due to shear slip, all of which depend on the mechanical properties of the joints. Only pertinent details of the fluid flow and mechanical deformation model are presented here. For a complete mathematical description, the reader is referred to the 3DEC documentation [40].

Formulation of the Permeability Tensor

The permeability tensor is calculated by extending the method proposed by [30] from two-dimensional to three-dimensional space. Darcy's law, which forms the basis of this calculation, can be expressed as follows:

$$\mathbf{Q} = \frac{\mathbf{KA}\Delta\mathbf{P}}{\mu L} \quad (11)$$

Here $\mathbf{Q}$ (m^3/s) represents the flow rate matrix, $\mathbf{K}$ (m^2) is the permeability tensor, A (m^2) is the cross-sectional area, $\Delta\mathbf{P}$ (Pa) is the pressure drawdown matrix, and μ is the fluid's dynamic viscosity (Pa s), and L is the length along the fluid's flow path (m). For orthogonal pressure gradients, the above equation can be expressed as follows:

$$\begin{bmatrix} q_{xx} & q_{xy} & q_{xz} \\ q_{yx} & q_{yy} & q_{yz} \\ q_{zx} & q_{zy} & q_{zz} \end{bmatrix} = \frac{A}{\mu L} \cdot \begin{bmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{bmatrix} \times \begin{bmatrix} \Delta p_{xx} & 0 & 0 \\ 0 & \Delta p_{yy} & 0 \\ 0 & 0 & \Delta p_{zz} \end{bmatrix} \quad (12)$$

$$\begin{bmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{bmatrix} = \frac{\mu L}{A} \cdot \begin{bmatrix} q_{xx} & q_{xy} & q_{xz} \\ q_{yx} & q_{yy} & q_{yz} \\ q_{zx} & q_{zy} & q_{zz} \end{bmatrix} \times \begin{bmatrix} \frac{1}{\Delta p_{xx}} & 0 & 0 \\ 0 & \frac{1}{\Delta p_{yy}} & 0 \\ 0 & 0 & \frac{1}{\Delta p_{zz}} \end{bmatrix} \quad (13)$$

Furthermore, principal permeabilities were determined by rotating the permeability matrix within a three-dimensional Cartesian coordinate system. The principal permeability directions correspond to the eigenvectors, and the respective eigenvalues represent their magnitudes. Mathematically, this can be expressed as follows:

$$\mathbf{K} = \mathbf{R} \mathbf{K}_p \mathbf{R}^{-1} \quad (14)$$

where $\mathbf{K}$ is the permeability matrix, $\mathbf{R}$ is a 3×3 matrix with each column representing an eigenvector, and $\mathbf{K}_p$ is the diagonal matrix of eigenvalues, with each diagonal element representing a principal permeability component. The equation can be expanded as follows:

$$\begin{bmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{bmatrix} = \mathbf{R} \times \begin{bmatrix} k_{\max} & 0 & 0 \\ 0 & k_{\text{med}} & 0 \\ 0 & 0 & k_{\min} \end{bmatrix} \times \mathbf{R}^{-1} \quad (15)$$

where $k_{\max}$, k_{med} , and $k_{\min}$ are the principal components of the permeability tensor in descending order.

Numerical Simulations and Verification

Model

The model comprises a cuboid with dimensions of 1 m, defined within the Cartesian coordinate system by the corners (0, 0, 0) and (1, 1, 1). Subsequently, this cuboid is partitioned into multiple blocks using predefined orientations of joint planes. Each block is then discretized into zones, to which material properties and constitutive laws were

assigned, rendering them capable of deformation. Similarly, contact laws governing interactions between blocks at joint planes were established, with specific joint properties allocated to each joint. As discussed previously, the deformation behavior of the blocks is governed by the Mohr–Coulomb model, while the Mohr–Coulomb joint model governs the interactions at joint planes. For a comprehensive mathematical exposition of these models, readers are referred to [44, 45].

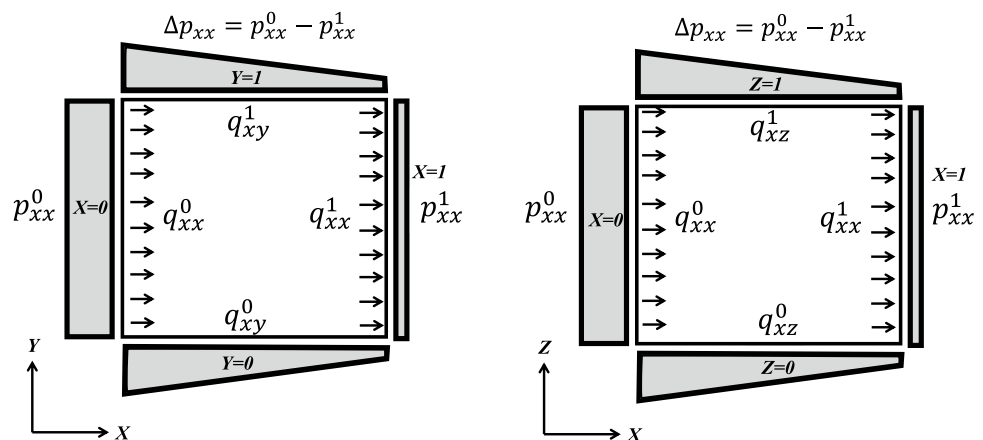
The numerical simulations were conducted in two phases, employing two distinct model configurations. Initially, simplified and idealized geometries were generated, for which analytical solutions for fluid flow and fracture deformation were established. These geometries serve as benchmarks for validating the outcomes of the current numerical simulations. Following successful verification, a digital representation of a geological columnar rock mass is constructed. Subsequently, three distinct stress loading scenarios were applied to explore the evolution of permeability under pol-axial stress conditions.

Boundary Conditions

All boundaries within the model allow fluid to pass through. To assess the flow rates of the first row of the discharge matrix (see Eq. 12), a pressure gradient (ΔP) is imposed along the X direction. The $X = 0$ face of the model is subjected to a pressure of p_{xx}^0 , while the $X = 1$ face experiences a pressure of p_{xx}^1 . Along the boundary planes (specifically, $Y = 0, Y = 1, Z = 0$, and $Z = 1$), a constant pressure gradient is employed (see Fig. 3). Constant stress boundary conditions are then enforced on each face, granting flexibility for the face walls to adjust according to the forces acting upon them. These stress boundary conditions were tailored to accommodate the specific stresses applied.

In this study, mechanical stresses and fluid flow are simulated in a sequential manner. Specifically, the model is initially brought to the desired stress state by gradually

Fig. 3 A schematic diagram depicting pressure gradients and discharge components in a model. Here the fluid pressure p_{xx}^1 is applied on the left face ($X = 0$), and the pressure p_{xx}^0 is applied on the right face ($X = 1$). A linear pressure gradient Δp_{xx} is applied along the $Y = 0, Y = 1, Z = 0$, and $Z = 1$ planes. The fluid discharge is measured perpendicular to all the six faces of the cube to construct the fluid discharge matrix (refer to Sect. 3.1.2)



applying incremental stresses. During this phase, no fluid pressure gradient is applied, and the system is maintained in a drained condition. Once the target stress state is attained, a constant fluid pressure gradient is introduced, and the fluid flow is subsequently simulated. At this stage, the mechanical solver is deactivated, meaning that no additional deformation occurs due to the fluid flow. The current study addresses a fluid-only simulation, or, more precisely, an uncoupled problem, wherein it is assumed that fluid pressure does not influence joint deformation. This assumption is justifiable given that the fluid pressure applied at the hydraulic head is 1 MPa, which is considerably lower than the applied principal stress range of 20–100 MPa.

Calculation of Fluid Discharge Components

Pressure gradients were applied in the X , Y , and Z directions, independently. To conduct the flow simulations, water, treated as a single-phase fluid, has been utilized. The relevant fluid properties are detailed in the accompanying Table 1. A schematic diagram depicting the pressure gradients and fluid discharge components is shown in Fig. 3. In each case, the fluid discharge is recorded on all six faces. For instance, when the pressure gradient ΔP_{xx} is applied along the X -direction, the flow discharge is measured on the upstream ($X = 0$) and downstream ($X = 1$) faces along the X -direction, as well as on the $Y = 0$, $Y = 1$, $Z = 0$, and $Z = 1$ faces, which are perpendicular to the flow direction

(see Fig. 3). The simulation runs until a steady-state flow is achieved, meaning that the discharge at two opposite faces is close within a predefined tolerance. The flow discharge component in each direction is calculated by averaging the discharges on two faces along that direction. This approach ensures robust measurement of fluid discharge by considering the influence of applied pressure gradients in all three spatial directions and by using averaged discharge values to account for potential variations across the measured faces. In general, diagonal elements of the discharge matrix can be calculated as follows (see Eq. 12):

$$q_{ii} = \frac{q_{ii}^0 + q_{ii}^1}{2}; \quad i = x, y, z \quad (16)$$

where Δp is applied along the i direction, and the discharge is measured on the plane orthogonal to the i direction. The convention used in the present study is that the first letter indicates the direction of the pressure gradient, while the second subscript denotes the direction of the face from which fluid discharge is measured. The superscripts 0 and 1 denote pairs of opposite faces along the X , Y , and Z directions. The off-diagonal components of the discharge matrix can be calculated as follows:

$$q_{ij} = q_{ji} = \frac{q_{ij}^0 + q_{ij}^1 + q_{ji}^0 + q_{ji}^1}{4}; \quad i \neq j; \quad i = x, y, z \quad \text{and} \quad j = x, y, z \quad (17)$$

where Δp is applied along i , and flow is measured in a plane orthogonal to the j direction, and Δp is applied along j , and flow is measured in a plane orthogonal to the i direction.

Table 1 Parameter values

Parameter	Value
Aperture initial	10^{-3} m
Aperture residual	10^{-5} m
Aperture maximum	10^{-2} m
Joint normal stiffness	100×10^9 Pa/m
Joint normal stiffness maximum	926×10^9 Pa/m
Joint shear stiffness	30×10^9 Pa/m
Joint stiffness exponent (normal and shear)	0.05
Joint friction	30°
Residual joint friction	22.5°
Joint cohesion	10×10^6 Pa
Residual joint cohesion	7.5×10^6 Pa
Matrix density	2500 kg/m^3
Matrix bulk modulus	84.5×10^9 Pa
Matrix shear modulus	60×10^9 Pa
Matrix friction	40°
Matrix cohesion	40×10^6 Pa
Fluid bulk modulus	2×10^9 Pa
Fluid density	1000 kg/m^3
Fluid viscosity	0.001 Pa s

Verification of Numerical Modeling Results

Fracture deformation with the applied normal stress and shear stress is verified on a single fracture, whereas permeability is validated on three categories of joint configurations.

Verification of Joint Deformation

In this scenario, a single joint is horizontally cut through the middle of the block (refer to Fig. 4). The material and mechanical properties assigned to both the matrix material and the joints are detailed in Table 1. During the application of normal stress ranging from 0 to 200 MPa, both the aperture and stiffness were recorded and compared with the corresponding analytical Eqs. (1 and 2). The verification of fracture normal closure is illustrated in Fig. 4. Additionally, dilation and residual aperture changes were monitored under shear stress and compared with the analytical solutions, as illustrated in Fig. 5. The numerical results of fracture normal deformation, shear deformation, and dilation were compared with analytical solutions for

Fig. 4 **a** Comparison between 3DEC (numerical) and analytical results, verifying the aperture change (black color) and joint normal stiffness (red color) variations with applied normal stress using the governing equations. **b** A schematic illustrating the contact model that governs aperture change with tensile and compressive stresses on the joint (Color figure online)

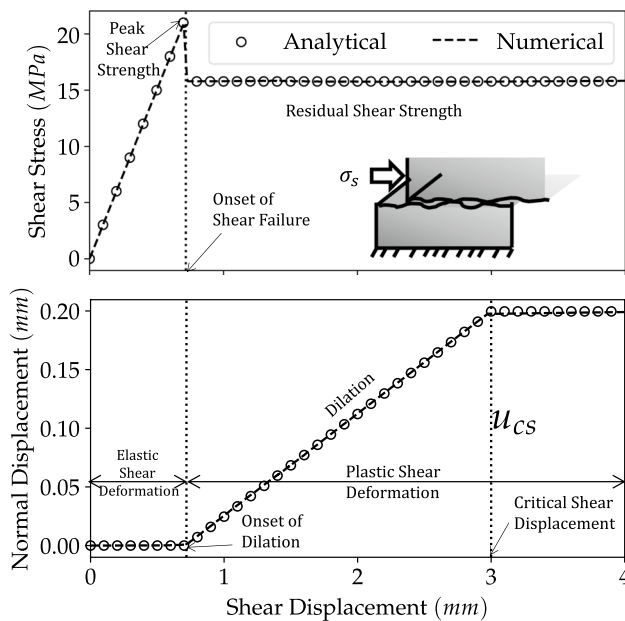
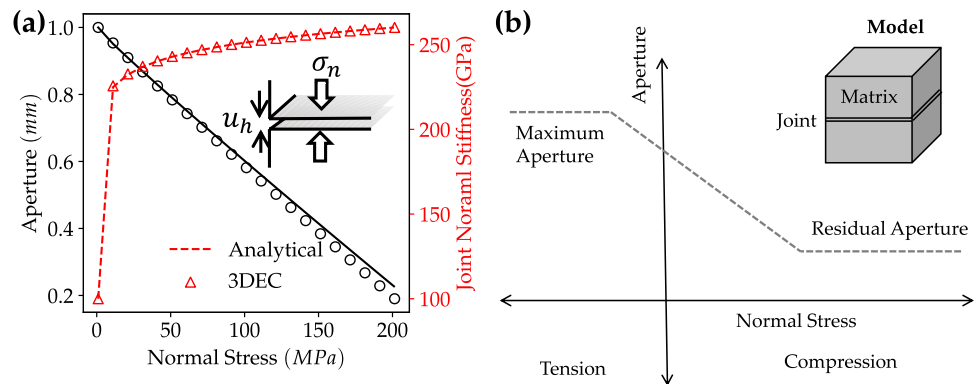


Fig. 5 Comparison of shear displacement and dilation variations between 3DEC (numerical) and analytical results. Top: Verification of joint shear displacement under applied shear stress. Bottom: Analysis of normal displacement and dilation variations with applied shear stress using analytical equations

a single joint, demonstrating good agreement. Fracture deformation manifests in several ways, including normal deformation, shear deformation, and dilation. Laboratory experiments consistently show that fracture apertures close or open under the application of normal stress, with the extent of this change dependent on the fracture's normal stiffness [46]. At high normal stresses, fractures are thought to exhibit a residual aperture, beyond which no further closure occurs. This residual aperture is influenced by the unmatched asperities of the fracture surfaces. Normal deformation is most significant at low normal stresses. Shear dilation of fractures occurs due to the overriding

asperities of rough fracture surfaces and is influenced by the degree of shear deformation.

Validation of Permeability

For the purpose of permeability verification, three joint geometries were constructed within the block: sheets, matchsticks, and cubes (refer to Fig. 6). The matrix properties and joint flow properties remain consistent across all three cases; the sole distinction lies in the arrangement of fractures within the model. Moreover, for each geometrical configuration, the fracture aperture systematically increases from 10^{-6} m to 10^{-3} m (refer to Table 2). As previously stated, the matrix is presumed impermeable. The material and mechanical properties assigned to both the matrix material and the joints are detailed in Table 1.

The permeability results were validated using simple geometries that have established analytical solutions. Specifically, we employed the three models described by [47]: sheets, matchsticks, and cubes as illustrated in Fig. 6 and Table 2. The analytical solutions for these models are provided in Table 2. For validation, a mean fracture spacing of $a = 0.25$ m was employed. The numerical results and the analytical results are presented in Fig. 6. These figures illustrate a close correspondence of 3DEC results with the analytical solutions. Additionally, it was observed that the principal permeability directions align with the geometry of the models, as anticipated. In the case of the sheets model, the Z-component of the permeability matrix is zero, resulting in a permeability tensor that can be visualized as a circle in the XY plane ($k_{zz} = 0$; $k_{xx} = k_{yy}$). Conversely, in the matchsticks model, the primary permeability aligns with the direction of the matchsticks or the Z-direction, while the other two principal components lie in the XY plane ($k_{zz} > k_{xx} = k_{yy}$). Similarly, in the cubes model, as expected, the permeability exhibits isotropic behavior ($k_{zz} = k_{xx} = k_{yy}$), represented visually as a sphere (see Fig. 6).

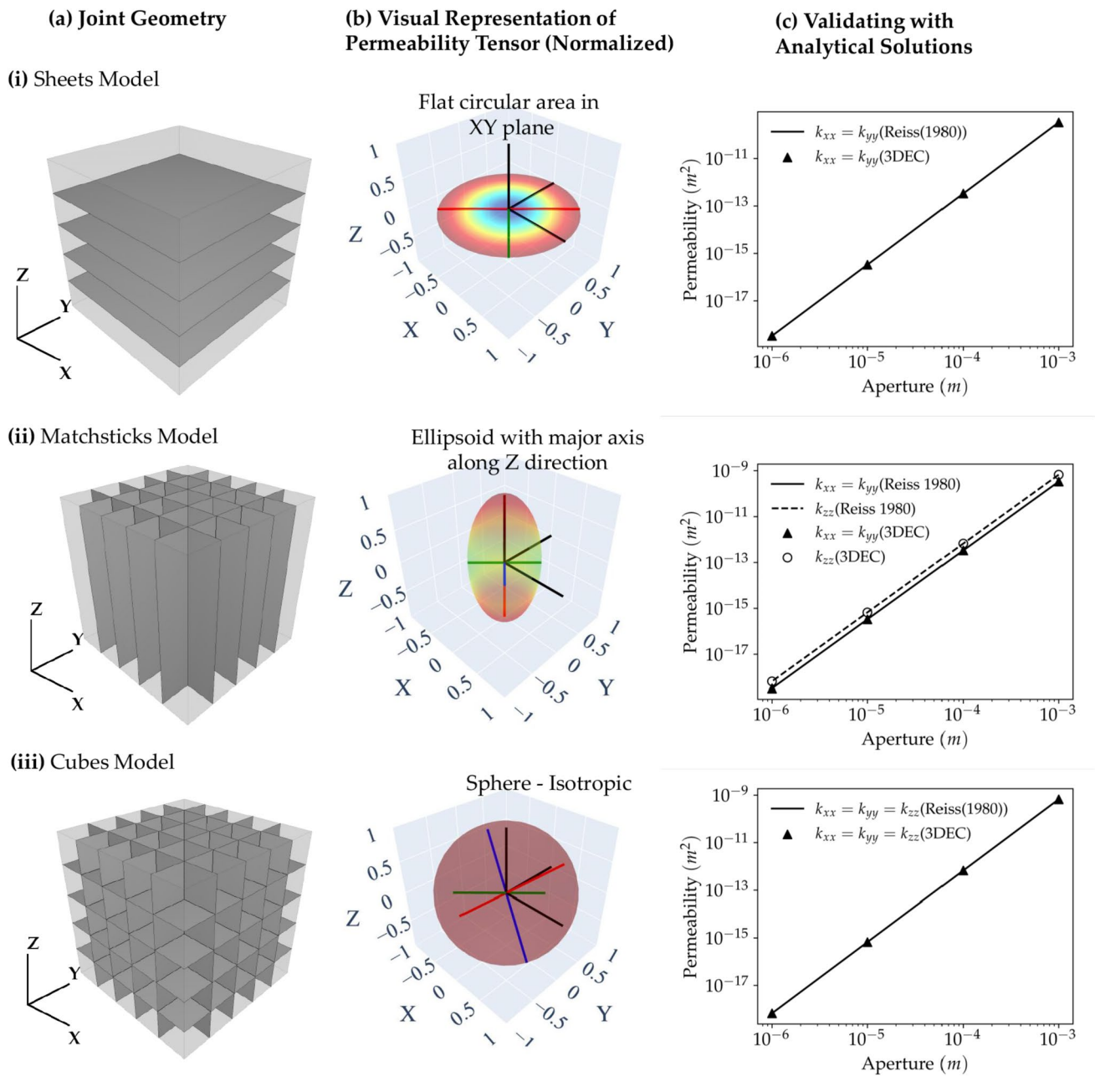


Fig. 6 Comparison of permeability components estimated using 3DEC and those obtained from analytical solutions, with a consistent fracture spacing ($a = 0.25$) across all models. Columns a, b, and c depict model geometry, visualization of the permeability tensor, and comparison between numerical and analytical results, respectively. Each row corresponds to a specific joint geometry: (i) sheets model, where $k_{xx} = k_{yy}$ and $k_{zz} = 0$; (ii) matchsticks model, with

$k_{xx} = k_{yy} \neq k_{zz}$; and (iii) cubes model, featuring an isotropic tensor, $k_{xx} = k_{yy} = k_{zz}$. In column b, the ellipsoid represents the normalized permeability tensor, with the maximum, intermediate, and minimum principal components depicted by red, green, and blue lines, respectively. The magnitude signifies the ratio of each permeability component to the maximum component

Table 2 Fractured rock model analytical equations for permeability

Model	Analytical equation for permeability	Aperture (b) (m)	Solution (m ²)
Sheets	$k_{xx} = k_{yy} = \frac{b^3}{12a}$	$a = 0.25$	
		10^{-3}	3.3333×10^{-10}
		10^{-4}	3.3333×10^{-13}
		10^{-5}	3.3333×10^{-16}
Matchsticks	$k_{xx} = k_{yy} = \frac{b^3}{12a}$ $k_{zz} = \frac{b^3}{6a}$	$a = 0.25$	
		10^{-3}	3.3333×10^{-10}
		10^{-4}	3.3333×10^{-13}
		10^{-5}	3.3333×10^{-16}
Cubes	$k_{xx} = k_{yy} = k_{zz} = \frac{b^3}{6a}$	$a = 0.25$	
		10^{-3}	6.6667×10^{-10}
		10^{-4}	6.6667×10^{-13}
		10^{-5}	6.6667×10^{-16}

Numerical Simulations of Columnar Jointed Rock Mass

Model and material properties

Columnar jointed rock mass (CJRM) is a geological structure commonly found in igneous rocks, formed through cooling and contraction processes. This structure consists primarily of polygonal columnar blocks with contacts described by tensile mode fractures. The columns are typically parallel and straight but can form curved sets due to deformation during or after formation. CJRM has significant applications in engineering fields, such as nuclear waste disposal and dam construction, etc. [35, 36].

The model used in this study comprises 50 columnar joint blocks within a cubic geometry with an edge length of 1 m. Block tessellations were generated using Neper open-source code [48] and imported into 3DEC code. The blocks

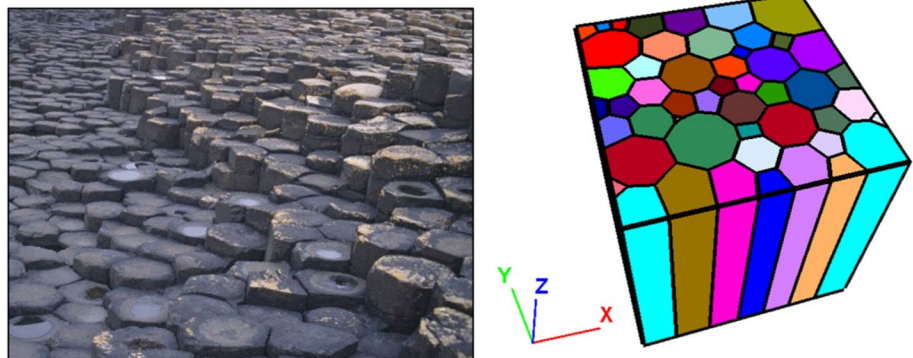
exhibit varied polygonal geometries and dimensions, with the columnar joint axis aligned along the Z-direction. Each block is modeled as isotropic, homogeneous, and impermeable (Fig. 7).

Joints are formed at the contacts between blocks and are assigned elastic properties and hydraulic apertures. Initial apertures of all joints, in the absence of applied stress, are assumed to be equal. As discussed in the previous section, the flow capacity of jointed rock primarily depends on joint aperture size. The aperture is affected by joint deformation, including normal closure, tensile opening, shear sliding, and dilation. Therefore, the elastic properties and constitutive laws assigned to the joints are critical model parameters influencing fluid flow in the rock. The model parameters are detailed in Table 1. For simplicity, it is assumed that the hydraulic and mechanical apertures are equal. Although fractures are assumed to be smooth and parallel-walled, fracture roughness significantly affects deformation and fluid flow behavior, potentially introducing intrinsic permeability anisotropy within a joint. Incorporating fracture roughness effects on fluid flow is beyond the scope of this research. It should be noted that the parameters chosen for this model aim to illustrate the permeability tensor and its behavior under different stress loading conditions and may not represent realistic in situ parameters of a specific field.

Table 3 Three stress loading scenarios considered in the present study

Scenario-I		Scenario-II		Scenario-III	
$\sigma_{xx} = \sigma_{yy} = \sigma_{zz}$		$\sigma_{yy} = 20 \text{ MPa}$, $\sigma_{zz} = 100 \text{ MPa}$		$\sigma_{xx} = 20 \text{ MPa}$, $\sigma_{zz} = 100 \text{ MPa}$	
$\sigma_{xx}(\text{MPa})$	σ_{yy}/σ_{xx}	$\sigma_{xx}(\text{MPa})$	σ_{yy}/σ_{xx}	$\sigma_{xx}(\text{MPa})$	σ_{yy}/σ_{xx}
20	1	20	1	20	1
40	1	40	2	40	2
60	1	60	3	60	3
80	1	80	4	80	4
100	1	100	5	100	5

Fig. 7 Columnar jointed rock (left) (Source: [49], used with permission). The 3DEC model (right) utilized in the present study



Stress Loading Scenarios

Three stress loading scenarios were considered: (I) Scenario-I: Isotropic stress loading, (II) Scenario-II: Increasing intermediate principal stress along the X-direction, and (III) Scenario-III: Increasing intermediate principal stress along the Y-direction. The principal stresses applied during the numerical simulation are detailed in Table 3. For each stress state, the model was brought to mechanical equilibrium by minimizing unbalanced forces. Subsequently, steady-state permeability measurements were taken.

Results and Discussion

Initial Permeability

The permeability tensor of the columnar joint set was calculated as described in the previous section. By applying pressure boundary conditions, single-phase steady-state water flow was simulated across the joint networks, as depicted in Fig. 3. To establish a consistent pressure gradient, constant hydraulic heads were applied to two opposing boundaries perpendicular to the flow direction, while the boundaries parallel to the flow direction were permeable. The mechanical and fluid properties used in the simulation are presented in Table 1. Initially, all fractures were assigned an equal hydraulic aperture of 10^{-3} m when $\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = 0$ MPa. The permeability tensorial values at the initial state are found as:

$$k = \begin{bmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{bmatrix} = \begin{bmatrix} 3.723 \times 10^{-10} & 2.902 \times 10^{-11} & 1.568 \times 10^{-15} \\ 2.902 \times 10^{-11} & 4.227 \times 10^{-10} & 5.988 \times 10^{-14} \\ 1.568 \times 10^{-15} & 5.988 \times 10^{-14} & 8.817 \times 10^{-10} \end{bmatrix} \quad (18)$$

The diagonalized permeability matrix is:

$$k_{\text{principal}} = \begin{bmatrix} k_{\text{max}} & 0 & 0 \\ 0 & k_{\text{med}} & 0 \\ 0 & 0 & k_{\text{min}} \end{bmatrix} = \begin{bmatrix} 8.817 & 0 & 0 \\ 0 & 4.359 & 0 \\ 0 & 0 & 3.590 \end{bmatrix} \times 10^{-10} \quad (19)$$

The corresponding eigenvector matrix, with each column representing an eigenvector (rounded to three decimal places), is:

$$R = \begin{bmatrix} 0.000 & -0.415 & 0.910 \\ 0.000 & -0.910 & -0.415 \\ 1.000 & 0.000 & 0.000 \end{bmatrix} \quad (20)$$

Using the min–max theorem of linear algebra, it can be demonstrated that k_{max} is always greater than or equal to the diagonal entries of the initial matrix k . For clarity, the principal components are denoted as $k_1 = k_{\text{max}}$, $k_2 = k_{\text{med}}$, and

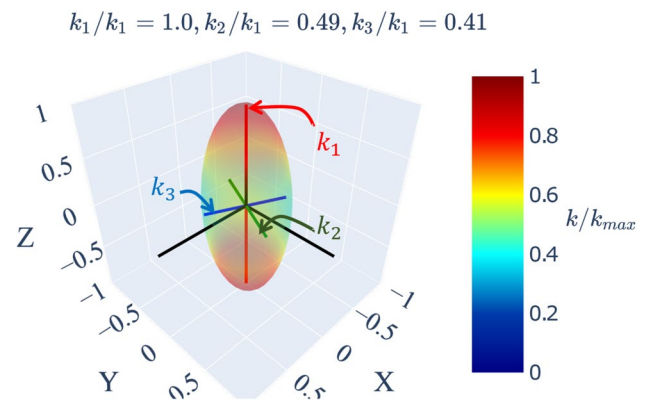


Fig. 8 Representation of the permeability tensor for the columnar jointed rock examined in this study. The maximum, intermediate, and minimum principal components are denoted by red, green, and blue lines, respectively. The length of each axis corresponds to the ratio of the respective permeability component to the major principal component. The major principal component aligns with the Z-direction, while the intermediate and minimum components lie within the XY plane

$k_3 = k_{\text{min}}$. The permeability tensor is depicted as an ellipsoid in Fig. 8. As observed from the $k_{\text{principal}}$ matrix, k_2 and k_3 are several orders of magnitude smaller than k_1 . This observation underscores the significance of treating permeability as a tensor and taking into account the directional dependence of fluid flow, rather than treating it as a scalar quantity.

As shown in Fig. 8, k_1 is aligned along the columnar axis direction (Z-direction), while k_2 and k_3 lie in the XY plane and are not oriented along the X or Y directions. This alignment indicates that the primary direction of fluid flow is parallel to the columnar joints, which has significant implications for understanding flow patterns in columnar jointed rock masses.

Scenario-I: Permeability Evolution—Isotropic Increase of Stresses

During the isotropic stress loading, all three principal stresses were simultaneously increased from 20 to 100 MPa in increments of 20 MPa. Under isotropic stress, all joints are uniformly compressed, resulting in hydraulic apertures smaller than the initial value of 10^{-3} m. This compression-induced reduction in aperture leads to a consistent decrease in the permeability of the entire rock mass under this loading condition, as depicted in Fig. 9. The principal components of the permeability tensor are shown in Fig. 10, while the normalized principal permeability values are depicted as ellipsoids in Fig. 11. It can be seen that during isotropic stress loading, the directions of the principal permeability components remain the same. However, the magnitudes of these values decrease at different rates, indicating an anisotropic evolution of the permeability.

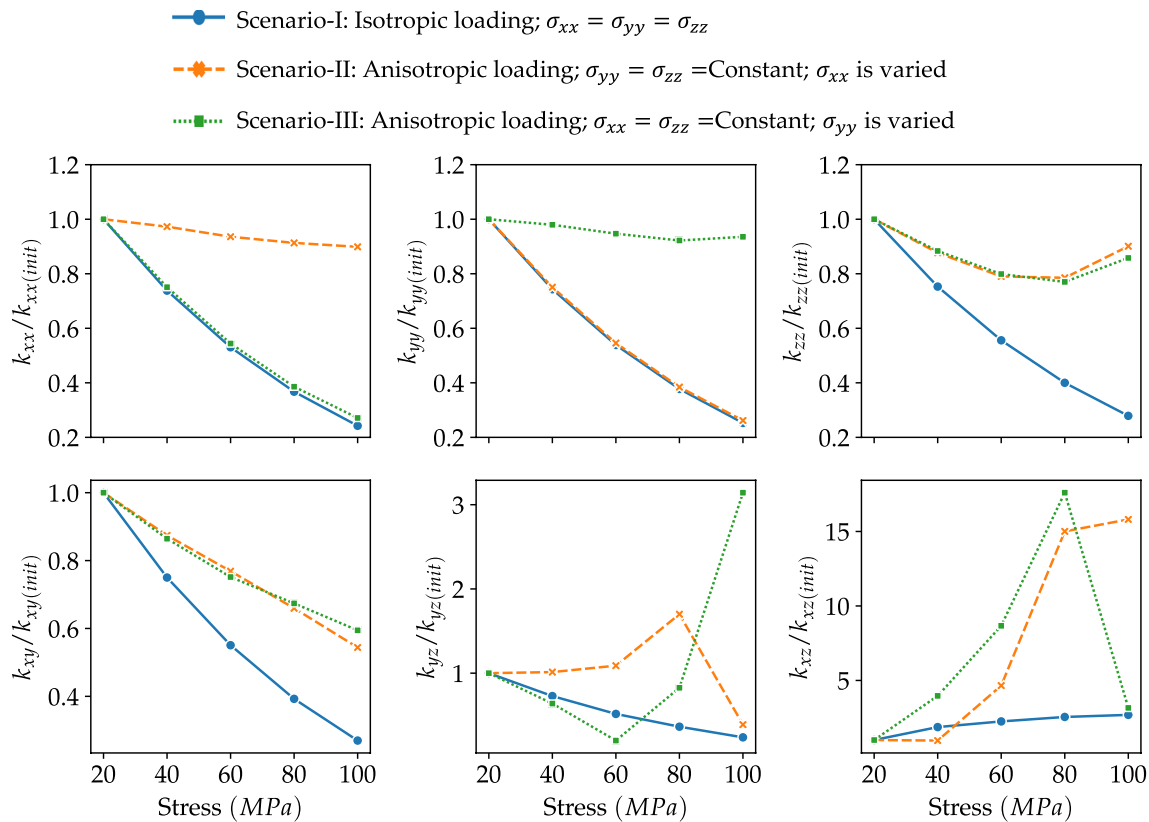


Fig. 9 Numerical simulation outcomes: The evolution of the six permeability components is charted alongside applied stress across three distinct scenarios, each denoted by a unique color and line style. The top row illustrates the diagonal components, while the bottom row

depicts the off-diagonal components. Each ratio reflects the permeability value at the corresponding stress level to the initial value (at 20 MPa)

Scenario-II: Permeability Evolution—Increase of intermediate principal stress along X-direction

In this scenario, the intermediate principal stress along the X-direction was varied from 20 to 100 MPa, while maintaining constant principal stresses of 20 MPa and 100 MPa along the Y and Z directions, respectively. When the jointed rock is subjected to anisotropic stress, a heterogeneous stress distribution develops, causing non-uniform compression of the joints. Each joint deforms based on the combined effects of compression and shear-induced dilation. The extent of compression and dilation in each fracture depends on the local distribution of normal and shear stresses and the movement constraints imposed by adjacent blocks. The relative angle between the joint plane and the principal stresses significantly influences the shear slip and dilation behavior of each joint.

In this scenario, the permeability tensor components exhibit mixed behavior, as shown in Fig. 9. The k_{yy} component decreases more rapidly than the k_{xx} and k_{zz} components. This is because joint planes aligned relatively perpendicular to the X-direction experience higher normal stress, leading to greater compression than joints oriented parallel to the

X-direction. Since the k_{yy} component is largely controlled by joints perpendicular to the X-direction, it undergoes a relatively rapid reduction under the applied stress. Notably, the k_{zz} component initially decreases but then begins to increase when σ_{xx} reaches 80 MPa (see Fig. 9). Beyond this stress level, critically oriented joints experience shear failure and dilation, thereby increasing the hydraulic aperture.

The principal components of the permeability tensor are shown in Fig. 10, while the normalized principal permeability values are depicted as ellipsoids in Fig. 11. The trends of the three principal permeability components (k_1 , k_2 , and k_3) also display mixed behavior (Fig. 10). The k_1 component initially decreases and then starts increasing after 80 MPa, while k_2 and k_3 monotonically decrease throughout the stress range. The k_1 component is predominantly influenced by the k_{zz} component and thus exhibits a similar trend. Unlike the isotropic case, where the principal permeability directions remain unchanged, the directions of the principal permeabilities change in this scenario. With increasing σ_{xx} , the k_1 component tends to align along the Z-direction, while the k_2 and k_3 components rotate within the XY plane, reorienting such that k_2 aligns along the Y-direction and k_3 along the X-direction (see Fig. 11). This reorientation occurs because

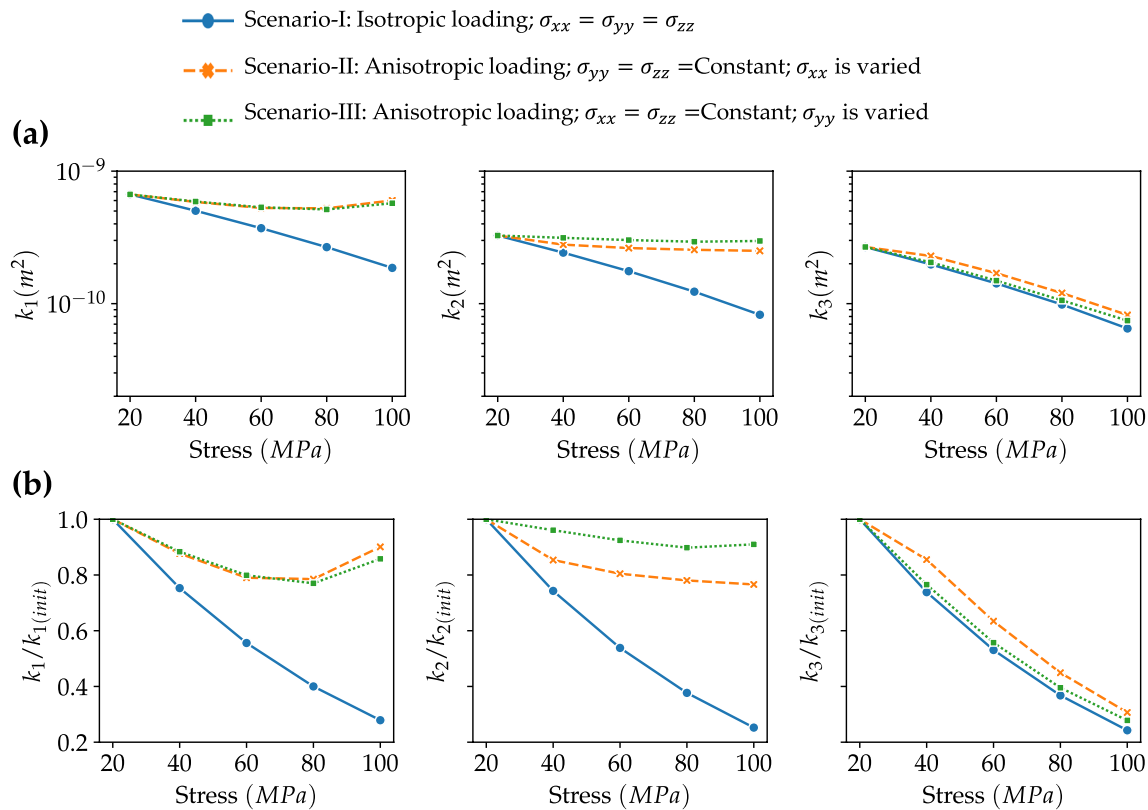


Fig. 10 Numerical simulation results: **a** progression of the three principal permeability components. **b** Ratios of the current permeability value to the initial value when the stress is 20 MPa. In each sce-

nario, the components are plotted against applied stress, distinguished by distinct colors and line styles

joints normal or subnormal to the X-direction experience relatively higher normal stress. These joints, being major conduits contributing to the k_{yy} component, reduce more rapidly compared to the k_{xx} component. Consequently, k_3 tends to decrease and reorient along the Y-direction, while k_2 reorients along the X-direction (see Fig. 11).

Scenario-III: Increase of intermediate principal stress along the Y-direction

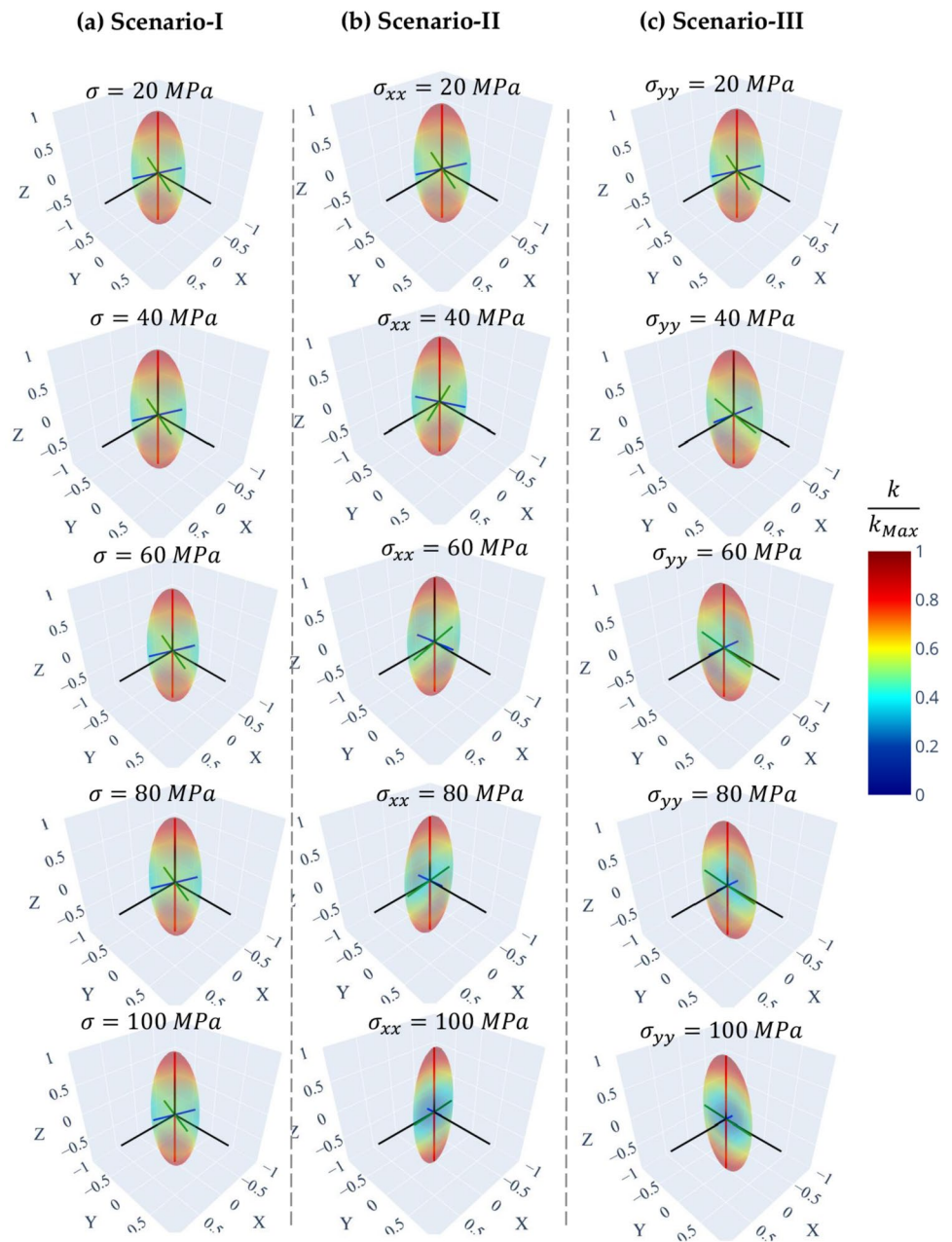
In this scenario, the intermediate principal stress σ_{yy} was elevated from 20 to 100 MPa, maintaining σ_{xx} at 20 MPa and σ_{zz} at 100 MPa. Consistent with the observations in Scenario II, variations in the permeability components were documented and depicted in Fig. 9. It can be noted that the permeability component k_{xx} diminished more rapidly compared to k_{yy} and k_{zz} . This pattern can be attributed to the enhanced normal compressive stress experienced by joint planes oriented perpendicular or nearly perpendicular to the Y-axis, which promotes the closure of apertures. Further analysis revealed dynamic behavior in the k_{zz} component, which initially decreases until σ_{yy} reaches 80 MPa, beyond which joint shear failure occurs, leading to dilation.

Figure 10 illustrates the principal elements of the permeability tensor, whereas in Fig. 11, the normalized principal permeability values are represented as ellipsoids. The behavior of the three principal permeability components (k_1 , k_2 , and k_3) also exhibited distinctive patterns as shown in figure. k_1 decreases initially and then increases post-80 MPa stress application. In contrast, k_2 and k_3 consistently decrease across the entire range of applied stress. As observed in Scenario II, the orientation of k_1 remains predominantly in the Z-direction, while k_2 and k_3 undergo rotational adjustments in the XY plane. Specifically, k_2 tends to align more with the Y-direction and k_3 with the X-direction, as illustrated in Fig. 11. This reorientation is primarily driven by the joint closure on planes that are perpendicular to the Y-direction.

Effect of Joint Stiffness

In the study of subsurface hydromechanical coupling problems, the role of joint normal stiffness (K_n) is critical. The impact of K_n under conditions of isotropic stress loading was studied, which corresponds to typical scenarios of pore pressure fluctuations in geological formations. As depicted

Fig. 11 Visual representation of numerical simulation results showing the principal components of the permeability tensor. Columns a, b, and c correspond to Scenario-I, Scenario-II, and Scenario-III, respectively, with each row indicating increasing stress levels (see Table 3). The maximum, intermediate, and minimum principal components are depicted by red, green, and blue lines, respectively. The length of each axis represents the ratio of the respective permeability component to the major principal component. In Scenario-I, the direction of the three principal components remains unchanged with increasing stress. In Scenario-II, the permeability tensor undergoes reorientation, with the intermediate component ($k_{intermediate}$) aligning with the Y-axis and the minimum component ($k_{minimum}$) decreasing further. In Scenario-III, the permeability tensor similarly reorients, with the intermediate component aligning with the X-axis and the minimum component decreasing further



in Fig. 12, an increase in K_n necessitates a higher effective stress to achieve fracture closure. It is observed that not all joints may close completely within the applied stress range for a given K_n , resulting in a residual aperture. Under such conditions, the permeability components reach a state of saturation, whereby further increases in applied stress do not lead to additional reductions in permeability.

Turning to joint shear stiffness (K_s), this parameter influences the ability of a joint to undergo shear deformation

and, ultimately, shear failure. Shear slip at the joint interfaces leads to dilation and the comminution of fracture wall asperities, thereby increasing the fracture aperture. Figure 13 illustrates the permeability variations across different values of K_s . It is evident that under isotropic stress conditions, which typically do not induce shear slip, the permeability remains unaffected by variations in K_s . This uniformity is due to the absence of shear deformation, with all joints experiencing uniform compressive stress (Fig. 13).

Fig. 12 Figure illustrating the sensitivity of the principal permeability component to stress at various initial joint normal stiffness values under isotropic stress loading. As joint normal stiffness increases, the evolution of permeability with stress becomes less rapid

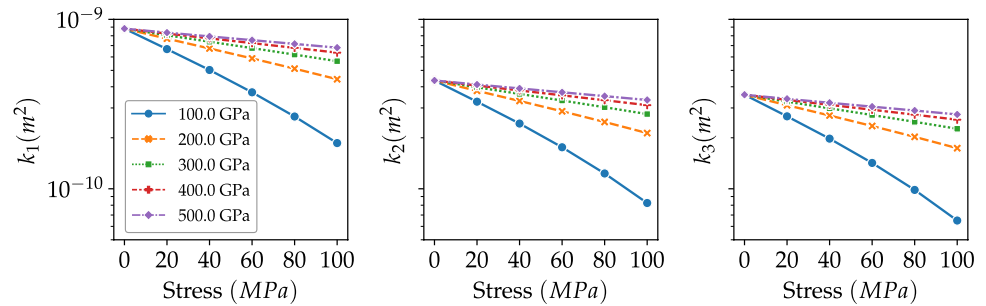
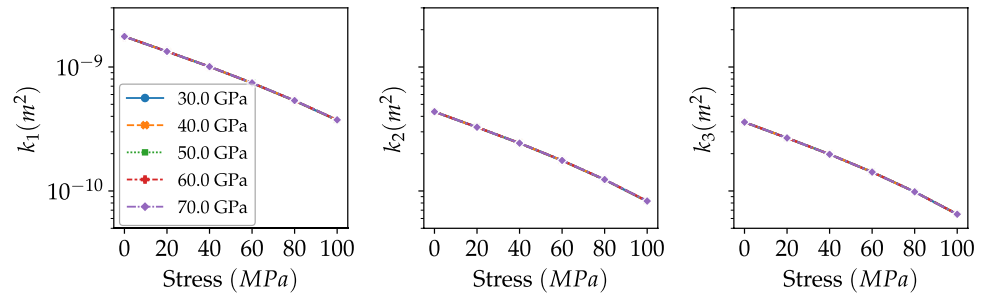


Fig. 13 Figure illustrating the sensitivity of the principal permeability component to stress at various initial joint shear stiffness values under isotropic stress loading. The evolution of permeability with stress is the same regardless of the shear stiffness



Empirical Model

In the present analysis, numerical simulation outcomes derived under isotropic stress loading were integrated with existing empirical models to ascertain stress-dependent permeability characteristics. The empirical approaches predominantly employed are the exponential and power-law models. The exponential model can be expressed as follows [21]:

$$k_{\sigma} = k_i e^{-\alpha(\sigma_e - \sigma_i)} \quad (21)$$

where k_{σ} represents the permeability at the effective stress σ_e , k_i is the initial permeability at σ_i , and α denotes a stress-sensitive coefficient (in MPa^{-1}), indicating the rock's sensitivity to applied stress with higher α values reflecting steeper decreases in permeability. Further studies refined this model for porous fractured rock by modeling α as a nonlinear function of fracture/pore compressibility and effective [22].

Alternatively, the power-law model can be expressed as follows [50]:

$$k_{\sigma} = k_i (\sigma_e - \sigma_i)^{-b} \quad (22)$$

Here b is a material constant. While the exponential model effectively fits data across various fields [22], certain studies suggest that the power-law model provides a superior fit for specific materials [51, 52]. In this study of isotropic stress loading—a process inherently linked to pore pressure changes due to fluid depletion or injection—it was observed that the exponential model aptly captures the permeability evolution, as depicted in Fig. 14. This model is particularly accurate across all principal permeability components, with

the fitting parameter α indicating the rate at which permeability diminishes under external confining stress. Notably, permeability components k_3 and k_2 decrease more rapidly than k_1 .

The relationship between the stress sensitivity coefficient α and the initial joint normal stiffness K_n was further investigated. A clear linear correlation was identified between the logarithm of α and the logarithm of K_n , as illustrated in Fig. 15. While we do not propose a more general empirical equation based on this observation at present due to limited data, incorporating additional experimental data could enable the development of a mathematical relationship between α and K_n in the future studies.

Limitations and Future Work

It is important to note that the permeability in this study is calculated using Darcy's law by measuring the bulk fluid flow through the fractured rock, considering its geometry, fluid properties, and pressure gradient. This calculated permeability represents the effective permeability of the fractured rock mass as a whole, which is different from the intrinsic permeability of the individual fracture sets or the permeability of the representative elementary volume (REV) of a homogenized rock. Understanding this distinction is crucial for accurately modeling and predicting fluid flow in geological formations with complex fracture networks. Further research is needed to validate these conclusions for different representative volumes and natural fracture systems.

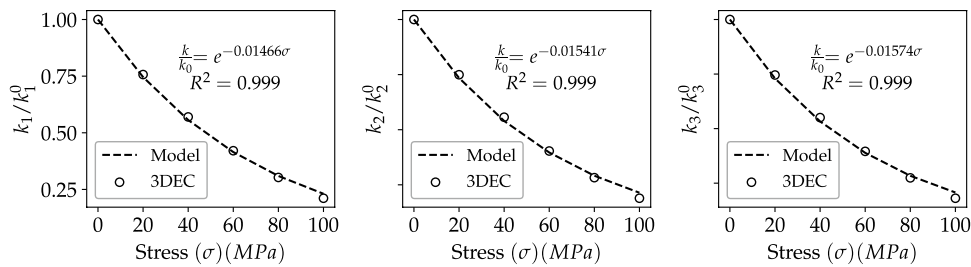


Fig. 14 Reduction of current to initial principal component permeability ratios under isotropic stress loading. Here $k_i^{(0)}$ represents the i^{th}

principal permeability component at zero stress. The markers denote results from 3DEC simulations, while the dashed line represents the exponential model fit to the data

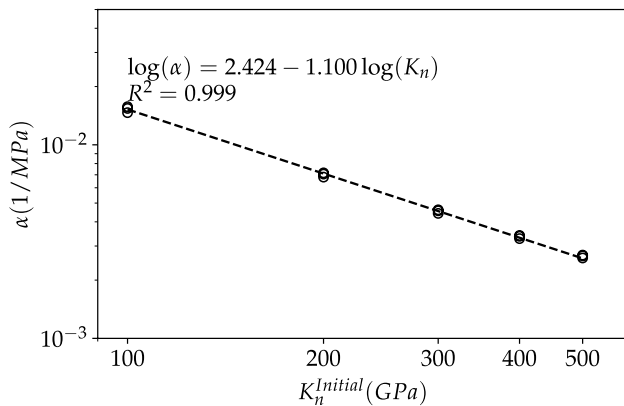


Fig. 15 Stress-sensitive coefficient (α) versus initial joint normal stiffness ($K_n^{Initial}$). Scattered points represent principal permeability components, and the dashed line depicts the empirical model fit

It should be noted that the current simulations have been validated using the analytical results of Reuss (1980) and not experimental or field data. While experimental validation would undoubtedly enhance the robustness of the model, setting up an experimental framework for CJRM is a significant challenge due to its complex nature. Future research involving experimental studies in this area would be highly valuable and could offer further insights into the phenomena. Moreover, although the current simulations pertain to a specific case of CJRM, they provide a solid framework for understanding permeability variations under stress changes. It is also worth noting that while different CJRM models may exhibit variations in their absolute permeability values, they are expected to demonstrate similar trends in permeability changes with respect to stress. In summary, while experimental or field validation would strengthen the findings, the current model offers a comprehensive approach to studying permeability variations in CJRM. The results should, therefore, be interpreted with these limitations taken into account.

It is to be noted that the numerical simulations performed are an uncoupled problem or fluid-only simulation. A fully coupled simulation would provide more realistic results.

The study focuses on the impact of stress on the permeability of jointed rock, particularly changes in fracture aperture. However, certain limitations exist, such as the assumption of smooth, parallel fracture walls, which overlooks the significant influence of fracture roughness on deformation and fluid flow behavior. Additionally, the study does not consider the potential for fractures to bend, propagate, or create new fractures. Further research is necessary to incorporate and comprehensively investigate these factors.

Conclusions

Modeling fluid flow in fractured rocks is a critical area of research due to its significance in many engineering applications. In this study, the three-dimensional blocky discrete element method was used to investigate the impact of intermediate principal stress on the permeability tensor of a three-dimensional columnar jointed rock. The key conclusions from the present research are as follows:

- The major principal component of the permeability tensor aligns with the columnar joint axis, while the other two components lie within the plane perpendicular to this axis. The median and minor principal permeability components may not always align with the principal stress directions.
- The findings highlight the importance of considering the intermediate principal stress in permeability evolution, an aspect often overlooked in the previous studies. Numerical simulations show that neglecting this effect can result in significant overestimations or underestimations of the permeability tensor components.
- Isotropic stress loading does not change the direction of the principal components of permeability, meaning the orientation of permeability remains constant. However, as the isotropic stress increases, the difference in permeability along different directions becomes more pronounced, leading to greater permeability anisotropy under higher isotropic stress conditions.

- Anisotropic stress changes alter both the magnitude and orientation of the permeability tensor components. Increased intermediate principal stress reorients the median and minor principal permeability components to align with the applied stress direction, while the orientation of the major component remains unchanged.
- The research underscores the necessity of characterizing permeability as a tensor rather than a scalar quantity. This distinction is crucial as three-dimensional subsurface stresses have distinct effects on each component of the permeability tensor, particularly in fractured rock systems with significant permeability anisotropy compared to porous rock systems.
- Reducing the normal stiffness of a joint increases its compliance under applied normal stress, causing the aperture to close more readily and accelerating permeability reduction. The shear stiffness of the joint does not significantly affect permeability reduction under isotropic stress loading.
- An exponential equation effectively represents the evolution of permeability under isotropic stresses, with the stress-sensitive coefficient showing a strong correlation with the initial joint normal stiffness. This correlation is valuable for modeling stress-induced changes in seepage characteristics within columnar joint rock formations.

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Author contributions Prasad Pothana helped in conceptualization; Prasad Pothana helped in methodology; Prasad Pothana helped in formal analysis and investigation; Prasad Pothana contributed to writing—original draft preparation; Kegang Ling and Vamegh Rasouli contributed to writing—review and editing; Kegang Ling and Vamegh Rasouli helped in funding acquisition; Kegang Ling and Vamegh Rasouli helped in resources; and Kegang Ling and Vamegh Rasouli worked in supervision.

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Data availability All the data necessary to support the findings of this study are included within the manuscript.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Consent for publication All authors have read and approved the final version of the manuscript for publication.

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